MIL-C-44303
30 June 1988

MILITARY SPECIFICATION

CLOTH, POROUS, LAMINATED FABRIC SYSTEM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This documents covers one type of porous, five-layer, laminated fabric system with a water repellent Woodland camouflaged-printed, outer shell fabric.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-P-1136 - Packaging of Coated (Plastic; Rubber) and Laminated Fabrics

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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MILITARY

MIL-C-44031 - Cloth, Camouflage Pattern, Woodland, Cotton and Nylon

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample. The laminated fabric system shall match the standard sample for color and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Materials,

3.2.1 Cloth, shell fabric (layer 1). The shell fabric cloth shall conform to class 2 of MIL-C-44031. The color shall be Woodland camouflage pattern.

3.2.2 Sorbent core fabric (layer 3). The sorbent core fabric shall be a melt-blown web of polypropylene microfibers loaded with physically entrapped active carbon granules in the 30 to 140 mesh range (see 6.5). The loaded melt-blown web shall be mechanically reinforced by inclusion of a system of larger fibers in or on the web. The active carbon shall be of a crush resistant type intended for gas and vapor absorption. The complete sorbent core fabric shall have a maximum weight of 7.5 ounces per square yard.

3.2.3 Adhesive (layers 2 and 4). The adhesive system for each adhesive layer (2 and 4) shall have a maximum weight of 0.4 ounce per square yard. Application of the adhesive system shall not expose active carbon in the core fabric (layer 3) to poisoning by vapors of organic solvents. The adhesive system shall be applied in a discontinuous pattern (see 6.4) which permits air permeability.

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3.2.4 Cloth, liner fabric (layer 5). The cloth for the liner shall be a tricot knit nylon knitted on a 28 gauge tricot machine using 40 denier nylon yarn. The color shall be black and show "good" fastness to water. The cloth shall weigh 1.8 minimum to 2.2 maximum, ounces per square yard. The yarns per inch shall be 37 minimum wales per inch and 35 minimum courses per inch. The bursting strength shall be 75 pounds minimum.

3.3 Construction of laminated fabric system. The laminated fabric system shall be composed of a five layer sequence as follows:

Layer 1 - Shell fabric
Layer 2 - Adhesive
Layer 3 - Sorbent core fabric
Layer 4 - Adhesive
Layer 5 - Liner fabric

The overlapping edges of core web (layer 3) shall not exceed 2 inches across the width. The width of each core layer segment, without overlapping, shall be not less than 16 inches.

3.4 Finished laminated fabric system. The finished laminated fabric system shall meet all the requirements in table I when tested as specified in 4.2.3.

TABLE I. Requirements of finished laminated fabric system

Characteristic	Requirement
Thickness (inches)	0.065 maximum
Peel strength between shell and liner (pounds)	2.0 minimum
Air permeability (cubic feet per minute per square foot)	6.0 minimum
Carbon tetrachloride absorption (mg/cm^2)	1.8 minimum

3.5 Width. The width of the finished laminated fabric system shall be 46 inches minimum.

3.6 Length and put-up. The finished laminated fabric system shall be put up in rolls 35 to 110 yards in length. No piece in a roll shall be less than 10 yards in length. No more than one splice per roll is permitted.

3.7 Workmanship. The finished laminated fabric system shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.1.1 Component and material certification. Unless otherwise specified, a certificate of compliance will be acceptable as evidence that the components and materials listed below conform to the specified requirements.

Requirement	Requirement paragraph
Sorbent core fabric:	
Weight 1/	3.2.2
Fibers	3.2.2
Active carbon	3.2.2
Adhesive system:	
Weight 1/	3.2.3
Application	3.2.3
Liner cloth material identification	3.2.3

1/ When tested by Method 5041 of FED-STD-121.

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4.2.2 End item examination.

4.2.2.1 Yard-by-yard examination. The laminated fabric system shall be examined for the defects listed in table II. The required yardage of each roll shall be examined on one side only, however, the side shall be alternated so that alternate rolls are examined on the face and back, respectively. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise or lengthwise yard or fraction thereof in which it occurs. The lot size shall be expressed in units of yards. The sample unit shall be 1 linear yard. The inspection level shall be III and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for total (major and minor combined) defects. The number of rolls from which the sample yardage is to be selected shall be as specified in table III. The sample yardage shall be apportioned equally among the selected rolls.

TABLE II. Yard-by-yard examination defects

Examine	Defect	Classification Major	Minor
Shell fabric	Color not as specified		X
	Hole, cut, mend, or weakened area:		
	- from 1/2 inch to 1 inch		X
	- larger than 1 inch	X	
	Creases or area of delamination	X	
Liner fabric	Color not as specified		X
	Hole, cut, tear, mend, run or dropped stitch, or abrasion mark resulting in weak area:		
	- from 1/2 inch to 1.0 inch in any direction		X
	- larger than 1 inch in any direction	X	
	Creases or area of delamination	X	
	Foreign matter		X
	Oil spot or stain		X
Width	Not as specified		X
General	Foreign matter		X
	Oil spot or stain		X
	Objectionable odor 1/	X	

1/ Odors of chemicals commonly used in laminating shall not be regarded as objectionable.

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TABLE III. Sample size

Lot size (yards	Sample size (rolls)	Acceptance number 2/
Up to and including 1200 1/	3	0
1201 to and including 3200	5	0
3201 to and including 10,000	8	0
10,001 to and including 35,000	13	0
35,001 to and including 150,000	20	1
150,001 and over	32	2

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

2/ Applicable to length examination defects only (see 4.2.2.2).

4.2.2.2 Length examination. Each roll of cloth used in the yard-by-yard examination shall be examined for the defects listed below. If the total number of defects in the sample rolls exceeds the applicable acceptance number specified in table III or if the total of the actual lengths of the sample rolls is less than the total of the lengths marked on the roll tickets, the lot shall be rejected.

Any roll (gross length) less than 35 or more than 110 yards
Any roll (gross length) more than 2 yards less than gross length marked on ticket
Any piece less than 10 yards long
More than one splice per roll
Any roll containing two or more pieces unspliced
Width less than 46 inches

4.2.3 End item testing. The laminated fabric system shall be tested for the characteristics listed in table IV. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table IV shall be followed. The sample unit shall be 3 yards, full width, for contractor testing (i.e., thickness, peel strength and air permeability) and a separate 3 yards, full width, for submission to DPSC for carbon tetrachloride absorption testing. The sample size shall be as indicated below. The lot shall be unacceptable if the lot average for air permeability, any individual carbon tetrachloride specimen result, or if any sample unit test result for any other characteristic fails to meet the specified requirements. All test reports shall contain the individual values utilized in expressing the final results.

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Lot size (yards)	Sample size (No. of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 up to and including 150,000	5

TABLE IV. End item tests

Characteristic	Requirement paragraph	Test method
Thickness	Table I	5030
Peel strength	Table I	5951
Air permeability	Table I	5450
Carbon tetrachloride absorption	Table I 1/	4.3.1

1/ Testing shall be performed by the Government (see 6.7). Except in cases of dispute, test samples shall be from production dried materials. In case of dispute, samples shall be conditioned in an air circulating oven at 50° ± 2°C for 16 ± 0.5 hours and tested while in the conditioned state.

4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1136.

4.3 Methods of inspection.

4.3.1 Carbon tetrachloride absorption.

4.3.1.1 Apparatus and reagents.

4.3.1.1.1 Vapor generator. Any apparatus can be used that generates a carbon tetrachloride/nitrogen gas mixture which contains a concentration of 5 ± 0.5 mg of carbon tetrachloride per liter (see 6.6).

4.3.1.1.2 Diluent. Prepurified nitrogen shall be used as the diluent to pass the carbon tetrachloride vapors through the test cups containing the test specimens. The flow through each test cup shall be 1 ± 0.03 liter/minute. The rate of flow can be controlled by the use of either a rotometer or a critical orifice.

4.3.1.1.3 Constant temperature cabinet. The constant temperature cabinet shall be large enough to hold the system, as shown in figure 1, and shall be capable of maintaining a temperature of 90° ± 2°F.

4.3.1.1.4 Fabric sample cup. The fabric sample cup shall conform to the requirements shown in figure 2.

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4.3.1.1.5 Pyrolyzer. The pyrolyzer shall conform to the requirements shown in figure 3. The platinum pyrolyzer is used between the outlet of the sample cup and the inlet of the "break-point" indicator bubbler. The pyrolyzer is heated by applying 10 volts, A.C., across the terminals during the test.

4.3.1.1.6 Bubbler. The bubbler shall conform to the requirements shown in figure 4.

4.3.1.1.7 Schwartz-type drying tube with stop-cocks. The tube shall be filled with activated charcoal as specified in 4.3.1.1.8, loosely packed, as shown in figure 5. Glass wool, loosely packed, shall be placed on top of the activated carbon. The filled tubes shall be stored in a desiccator when not in use.

4.3.1.1.8 Charcoal, cocoanut, activated, 6 to 14 mesh.

4.3.1.1.9 Indicator solution. Dissolve 2.5 grams of potasium iodide in 1000 mL of deionized water. Mix 90 mL of this solution with 10 mL of a 1-percent, stable, starch solution. The starch/iodide solution is used in the indicator bubbler. Prepare a fresh starch/iodide solution daily.

4.3.1.1.10 Desiccator.

4.3.1.1.11 Analytical balance.

4.3.1.1.12 "C" type clamp.

4.3.1.1.13 Ice water bath.

4.3.1.2 Determination of carbon tetrachloride concentration.

4.3.1.2.1 Tare weight of drying tube. The Schwartz-type drying tube, filled with charcoal and prepared as shown in figure 5, shall be equilibrated to constant weight using the apparatus with a flow of 1 ± 0.03 liter/minute of nitrogen (see figure 1), no carbon tetrachloride. The Schwartz-type drying tube replaces the sample cup for this determination.

4.3.1.2.2 Determination of carbon tetrachloride concentration. Sample the carbon tetrachloride vapor from the generator manifold at a rate of 1 liter/minute through the tared Schwartz-type drying tubes (with stop-cocks) for a 15-minute period. Close stop-cocks, cut off the vacuum and disconnect the Schwartz-type drying tube. Allow the tube to come to thermal equilibrium for 10 minutes at ambient temperature. Open and close one stop-cock slowly to equilibrate pressure. Weigh the tube to the nearest milligram.

4.3.1.2.3 Calculation. The carbon tetrachloride concentration (mg/liter), is determined by dividing the weight pickup (in mg) by the sampling time (minutes), and flow rate (liters/minute). Determine the carbon tetrachloride concentration during each test run.

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4.3.1.3 Test specimen. The test specimen shall be a 13.6 ± 0.2 cm diameter piece of the finished fabric. For each 3-yard sample unit, cut three specimens equally spaced along left side, three specimens equally spaced along right side, and three specimens equally spaced along the center of the sample unit. Each specimen shall be tested.

4.3.1.4 Procedure. Assemble the complete apparatus as shown in figure 1. For each test, mount one layer of the material in the test cup, shell fabric side up. The up-side of the cup is the inlet side in respect to carbon tetrachloride flow. After placing the samples in the cup, clamp the cup halves together with a "C" type clamp. The clamp shall remain on the cup during the entire test. Place 10 mL of the indicator solution in each of the indicator bubblers and immerse the indicator bubblers in an ice water bath and keep immersed during the test. A white background shall be used behind the bubbler to facilitate color detection. Start the flow of carbon tetrachloride through the test sample and record the time for the first tinge of blue to appear in the indicator bubblers. This establishes the original recorded time as the true "break" time. This bubbler is replaced with a fresh bubbler which is run until the latest new bubbler "breaks" within a 2-minute period. If the second bubbler does not show the blue tinge, the test is continued until the second bubbler does show the blue tinge, and this additional time is added to the original break time to establish a new true "break" time. This bubbler is replaced with a fresh bubbler which is run until the latest bubbler "breaks" within a 2-minute period. The additional bubblers are required to confirm that a true "break" has been obtained. The sample cup shall be purged with air after running each sample to assure removal of any residual carbon tetrachloride.

4.3.1.5 Calculation. The carbon tetrachloride absorption shall be calculated as follows:

$$mg/cm^2 = \frac{T \times C}{A} \times F$$

Where: T = true break time (minutes)
C = mg carbon tetrachloride/liter
A = area of test specimen (cm^2)
F = flow rate (1 liter/minute)

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A.

5.1.1 Level A. The laminated fabric system shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1136, except that each roll shall be unit packed in two polyethylene bags in accordance with alternative unit packing cited in PPP-P-1136. Prior to or during closing, excess air within the bag or tubing shall be removed. The first bag or tubing around the roll of cloth shall be sealed before insertion into the second bag or tube.

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5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The cloth, shall be packed in accordance with the applicable requirements of PPP-P-1136.

5.3 Marking. In addition to any special marking required in the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1136.

6. NOTES

6.1 Intended use. The laminated fabric system cloth is intended for use in a chemical-protective wrap to protect medical patients from penetrating, toxic, military agents.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

6.4 Adhesive pattern. Care should be exercised in the design of the adhesive pattern in order to meet air permeability requirements specified in table I.

6.5 Warning. Prior to and after processing, active carbon should be protected from exposure to solvent vapors, vehicle exhaust, cigarette smoke and other contaminants which can reduce absorption efficiency.

6.6 Toxicity. Due to the toxicity of carbon tetrachloride, the effluent must be vented into a hood. The exit gases of the vacuum pump and manifold must be vented (see 4.3.1.1.1).

6.7 Address. Samples for testing should be addressed to Commander, Defense Personnel Support Center, ATTN: DPSC-T7L, 2800 South 20th Street, Philadelphia, PA 19101-8419.

6.8 Subject term (key word) listing.

Cloth
Fabric system
Laminate
Porous

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Custodians:

Army - GL
Navy - NU

Review activities:

Army - EA
DLA - CT

User activity:

Navy - MS

Preparing activity:

Army - GL

Project No. 8305-0199

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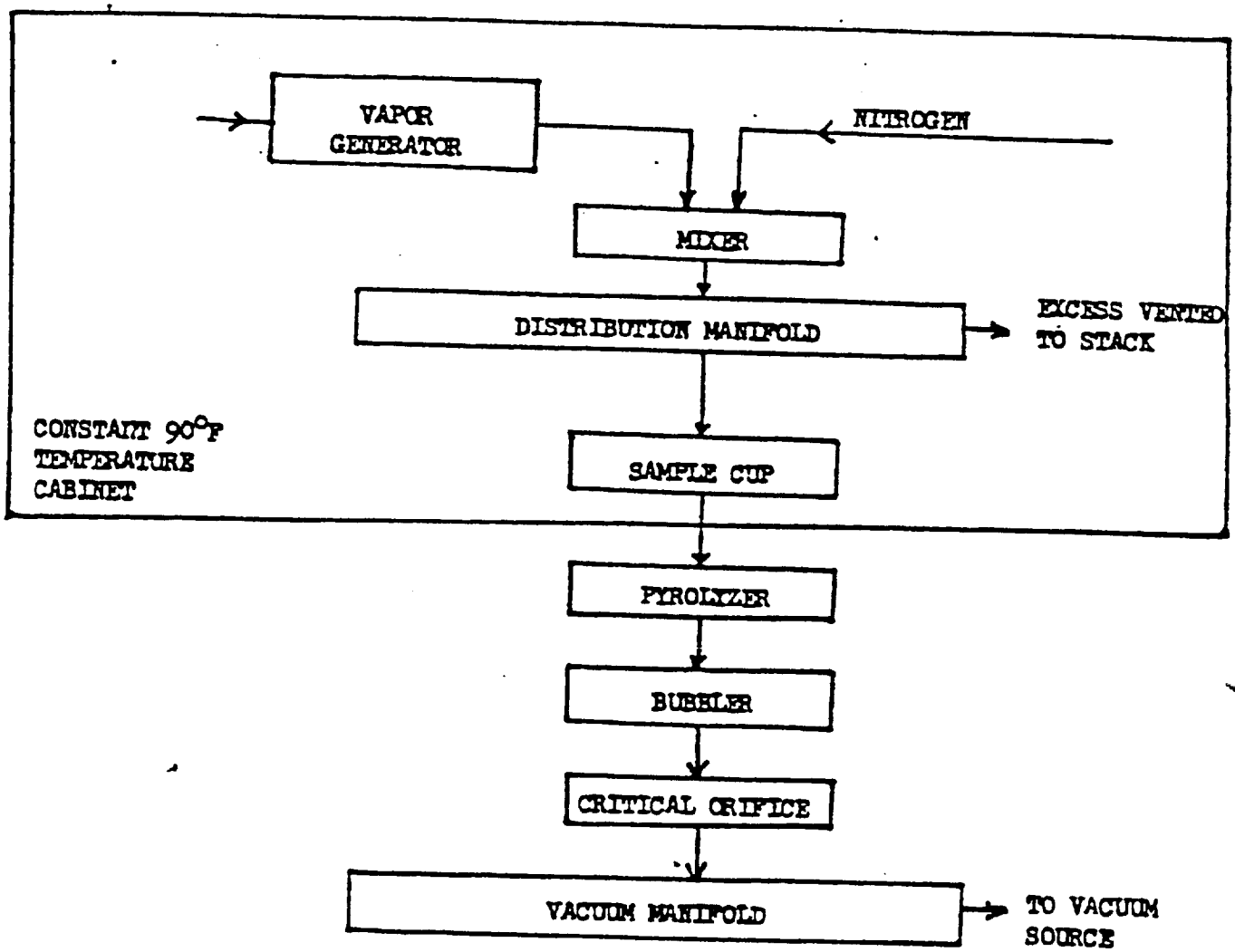
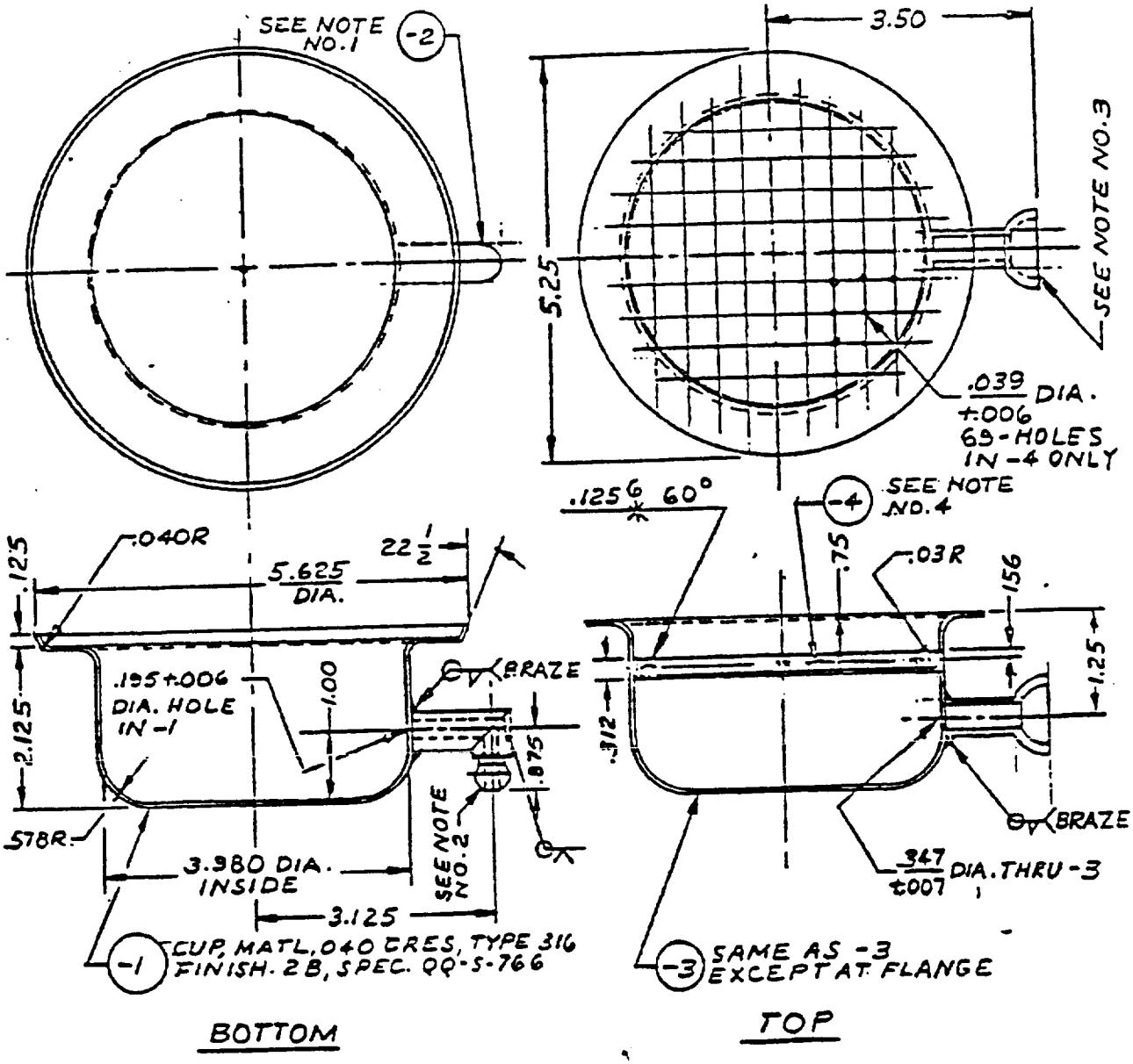


FIGURE 1

FLOW DIAGRAM OF VAPOR PENETRATION APPARATUS

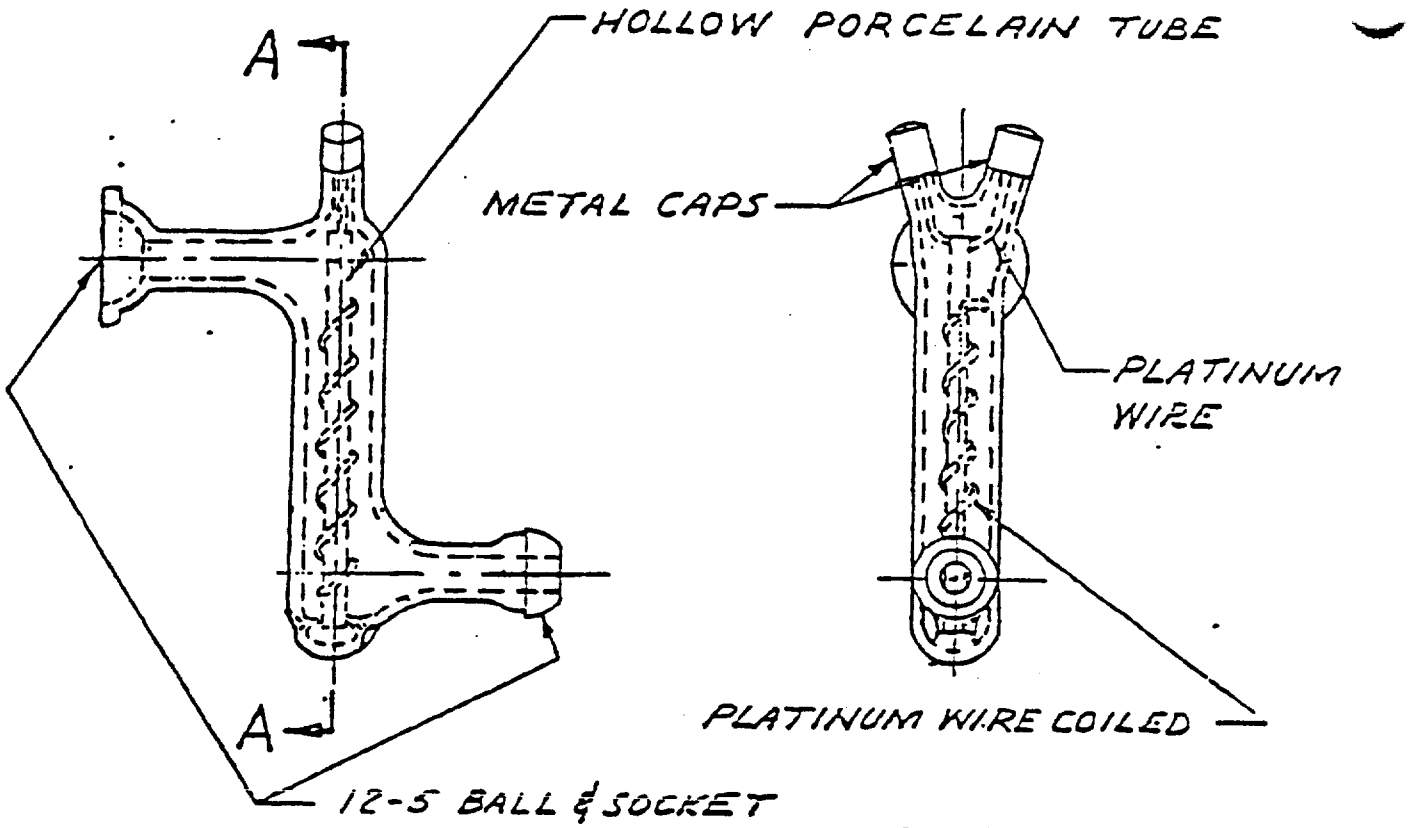
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NOTE:
1. 540-.010 O.D. x .190 +.006 I.D. CRES TYPE 316, SPEC. QQ-S-763.
2. GROUND JOINT, S, METAL BALL SIZE 12/5, TYPE 316, CRES. KONTES, TECH GLASS CO., VINELAND, N.J.
3. GROUND JOINT S, METAL SOCKET SIZE 18/9 TYPE 316 CRES. KONTES TECH GLASS CO., VINELAND, N.J.
4. MAT'L., .021 THICK, CRES, TYPE 316, FINISH 2B, SPEC QQ-S-766
5. REMOVE ALL BURRS & BREAK SHARP EDGES.

FIGURE 2. FABRIC SAMPLE CUP

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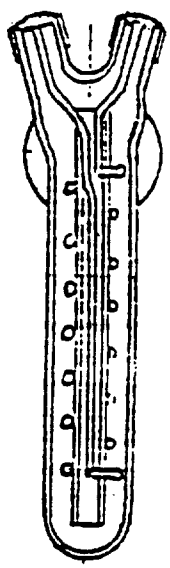


Materials:

(1) 18 gauge platinum wire used in leads.

(2) 30 gauge platinum wire used for heating element. Wire is 36 inches long and is wound around a 0.040 diameter wire to produce a coiled element. The 0.040 diameter wire is removed from the coiled wire. One inch of wire is left uncoiled on each end of the heating element to facilitate fusion to the leads.

(3) The coiled heating element is wrapped around a hollow alundum tube 1/8 inch O.D. and 1/6 inch I.D.



Scale 1/1

SECTION A-A

FIGURE 3
PYROLYZER

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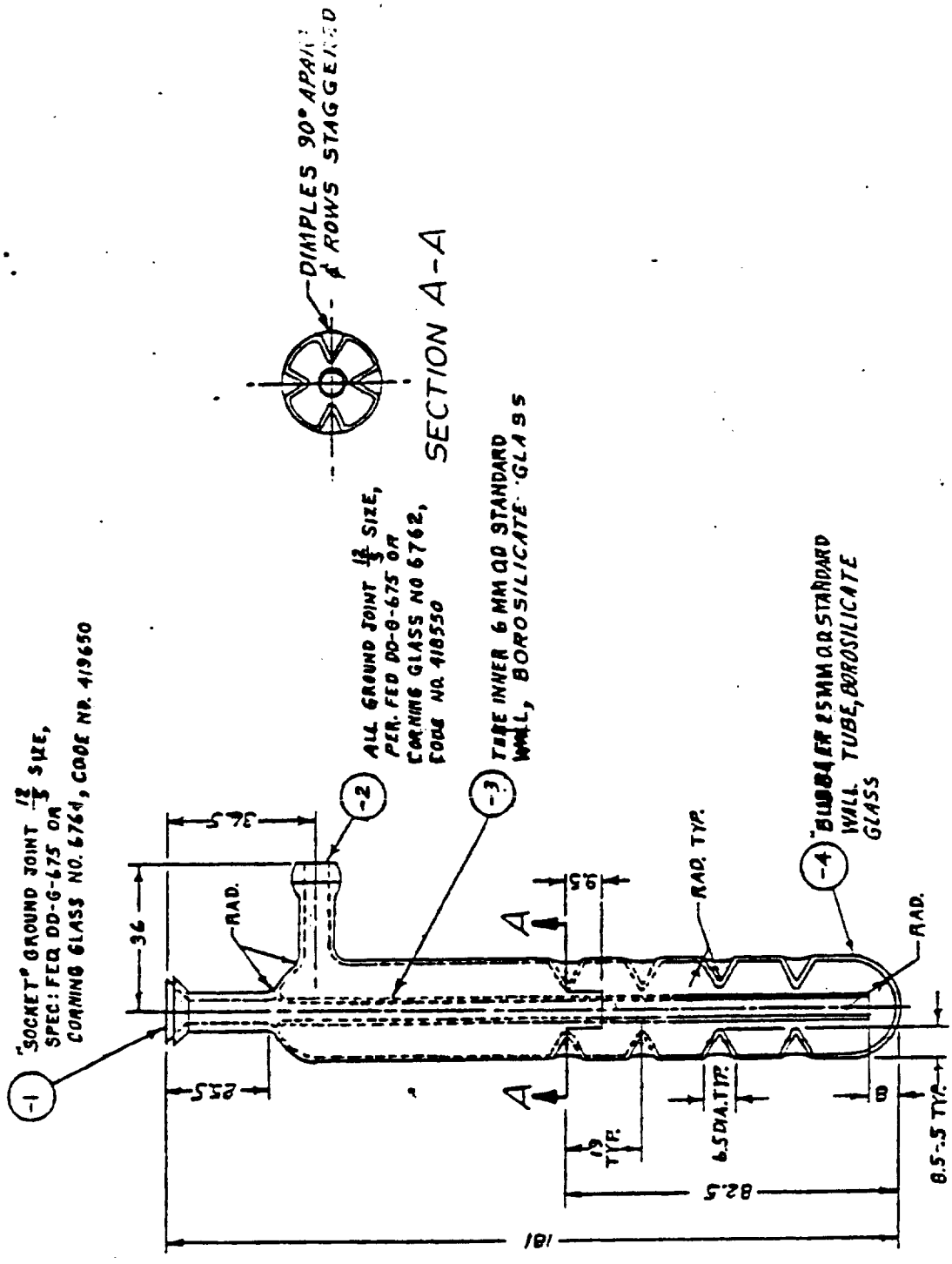


FIGURE 4. BUBBLER

NOTE

1/ ALL DIM. ARE IN MILLIMETERS.
TOL.: ± 1 MM

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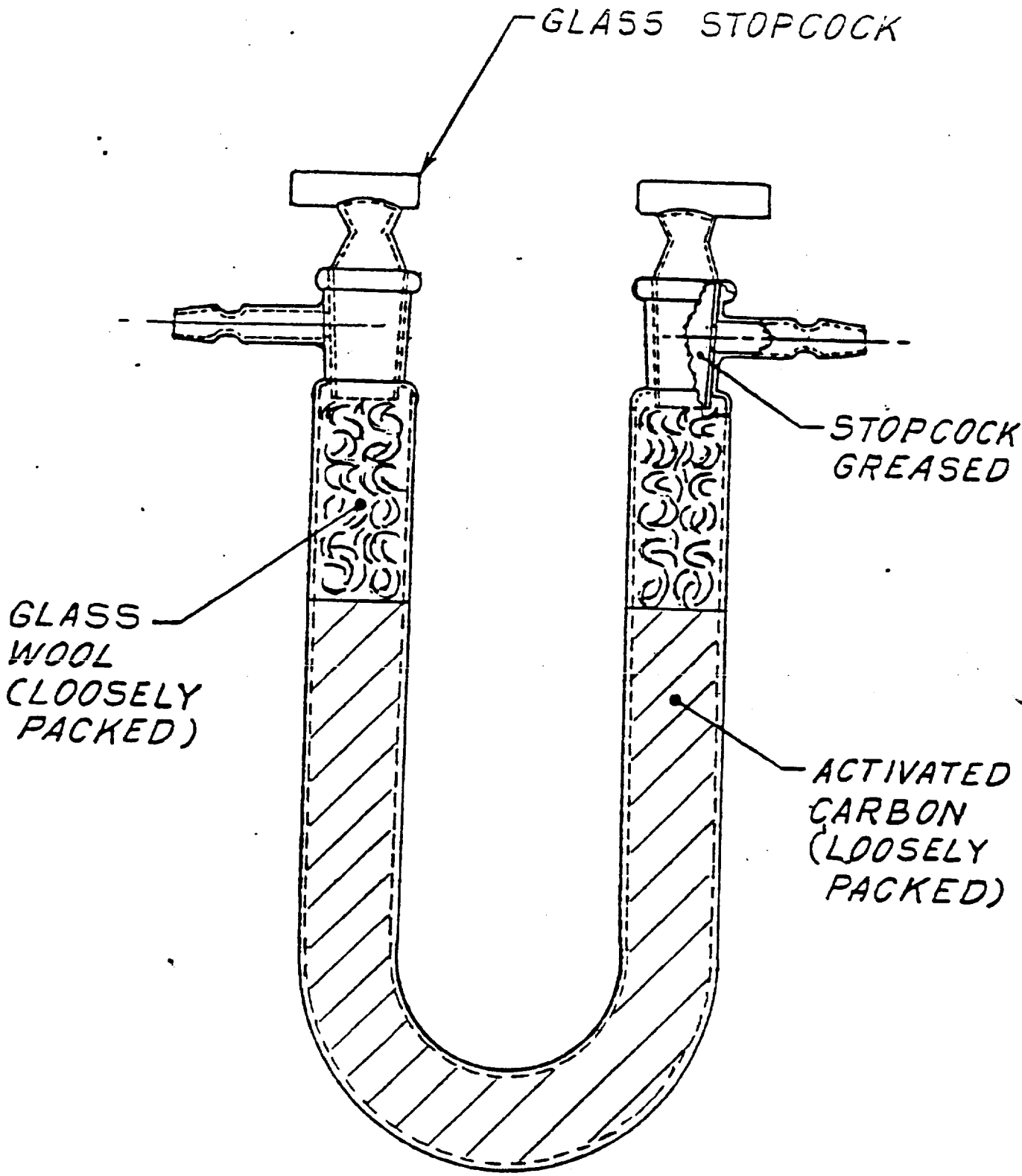


FIGURE 5.

SCHWARTZ TYPE DRYING TUBES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44303	Cloth, Porous, Laminated Fabric System

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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